Integral Transform

Fourier Transform

Fourier Series

Definition

For a periodic function $f(t)$, the Fourier series representation is given by

$$f(t) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{2\pi n}{T} t\right) + b_n \sin\left(\frac{2\pi n}{T} t\right) \right), \quad \text{for} - \pi \leq t \leq \pi$$

where

$$\begin{aligned} a_0 &= \frac{1}{T} \int_0^T f(t) dt \\ a_n &= \frac{2}{T} \int_0^T f(t) \cos\left(\frac{2\pi n}{T} t\right) dt \\ b_n &= \frac{2}{T} \int_0^T f(t) \sin\left(\frac{2\pi n}{T} t\right) dt \end{aligned}$$

The complex exponential form is given by

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{i \frac{2\pi n}{T} t}$$

where

$$c_n = \frac{1}{T} \int_0^T f(t) e^{-i \frac{2\pi n}{T} t} dt$$

If function is defined on $[-L, L]$, then we need linear transform $y = \frac{Lx}{\pi}$ coefficients are given by

$$\begin{aligned} F(y) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos ny + b_n \sin ny] \\ a_0 &= \frac{\int_{-\pi}^{\pi} F(y) dy}{\pi} = \frac{\int_{-L}^{L} f(x) \frac{\pi}{L} dx}{\pi} = \frac{\int_{-L}^{L} f(x) dx}{L} \\ a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} F(y) \cos(ny) dx = \frac{1}{L} \int_{-L}^{L} f(x) \cos(\frac{n\pi x}{L}) dx \\ b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} F(y) \sin(ny) dx = \frac{1}{L} \int_{-L}^{L} f(x) \sin(\frac{n\pi x}{L}) dx \end{aligned}$$

Fourier Transform

Definition

The Fourier Transform decomposes a function (usually a time-domain signal) into a sum (or integral) of sine and cosine waves of different frequencies.

$$\hat{f}(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$

The inverse Fourier Transform can recover the original signal from its frequency representation:

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(\omega) e^{i\omega t} d\omega$$

Properties

Linearity

The Fourier Transform of a linear combination of functions is the linear combination of the Fourier Transforms of those functions.

$$\mathcal{F}(c_1 f_1(t) + c_2 f_2(t)) = c_1 \hat{f}_1(\omega) + c_2 \hat{f}_2(\omega)$$

Time Shifting

Shifting a function in the time domain results in a phase shift in the frequency domain.

$$\mathcal{F}(f(t - t_0)) = e^{-i\omega t_0} \mathcal{F}(f(t))$$

Frequency Shifting

A shift in the frequency domain corresponds to a multiplication of the time-domain function by an exponential.

$$\mathcal{F}^{-1}(\hat{f}(\omega - \omega_0)) = f(t) e^{i\omega_0 t}$$

Differential Property

The Fourier Transform of the derivative of an function equals to $i\omega\hat{f}(\omega)$.

$$\mathcal{F}(f') = i\omega\mathcal{F}(f)$$

Integral Property

If we have a function such that $\int_{-\infty}^{\infty} f(t)\mathrm{d}t = 0$, then

$$\mathcal{F}\left(\int_{-\infty}^{t} f(t)dt\right) = \frac{1}{i\omega}\mathcal{F}(f)$$

If $\int_{-\infty}^{\infty} f(t)\mathrm{d}t \neq 0$, then

$$\mathcal{F}\left(\int_{-\infty}^{t} f(t)dt\right) = \frac{1}{i\omega}\mathcal{F}(f) + \pi\hat{f}(0)\delta(\omega)$$

Convolution Theorem

The Fourier Transform of the convolution of two functions is the pointwise product of their Fourier Transforms.

$$\mathcal{F}(f_1 * f_2) = \hat{f}_1 \cdot \hat{f}_2$$

Parseval's Theorem

This theorem states that the total energy of a signal (integrated over all time) is the same in both the time and frequency domains.

$$\int_{-\infty}^{\infty} f(t)^2\mathrm{d}t = \frac{1}{2\pi}\int_{-\infty}^{\infty} |\hat{f}(\omega)|^2\mathrm{d}\omega$$

In terms of Fourier Series, it can also be written as

$$\int_{-\pi}^{\pi} |f(x)|^2 dx = 2\pi \sum_{n=-\infty}^{\infty} |c_n|^2$$

$$c_n = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)e^{-inx}dx$$

Generally, for the Fourier Series

$$f(x) = a_0 + \sum_{n=1}^{\infty}\left(a_n \cos\frac{n\pi x}{L} + b_n \sin\frac{n\pi x}{L}\right)$$

the Parseval's theorem is given by

$$\frac{1}{2L}\int_{-L}^{L} f^2(x)dx = a_0^2 + \frac{1}{2}\sum_{n=1}^{\infty}\left(a_n^2 + b_n^2\right)$$

Laplace Transform

Fourier Transform requires a function satisfies the Dirichlet condition and is absolutely integrable over $(-\infty, \infty)$. However, many realistic function cannot meet the condition. For example, function with respect to time $(t > 0)$. For an arbitrary function $\phi(t)$, we can utilize properties of **unit step function** and **exponential decay function** to make it suitable for Fourier transform. $u(t)$ can converts the domain to $[0, \infty)$, and $e^{-\beta t}$ can make the function to be absolutely integrable. Therefore, consider the following function:

$$\phi(t)u(t)e^{-\beta t}, \quad \beta > 0$$

The Fourier transform always exists with suitable β. Do Fourier transform on it,we obtain the **Laplace Transform**.

Definition

Laplace transform converts integral and differential equations into algebraic equations. I's defined as:

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty f(t)e^{-st}dt$$

Laplace Transform Table

The Laplace transform table is shown as follows:

Function $ f(t)$	Laplace Transform $ F(s) = \mathcal{L}{f(t)}$
1	$\frac{1}{s}$, $s > 0$
$\frac{t^n}{n!}$	$\frac{1}{s^{n+1}}$, $n \geq 0$
e^{at}	$\frac{1}{s-a}$, $s > a$
te^{at}	$\frac{1}{(s-a)^2}$, $s > a$
$t^n e^{at}$	$\frac{n!}{(s-a)^{n+1}}$, $s > a$
$\sin(\omega t)$	$\frac{\omega}{s^2+\omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2+\omega^2}$
$e^{at}\sin(\omega t)$	$\frac{\omega}{(s-a)^2+\omega^2}$
$e^{at}\cos(\omega t)$	$\frac{s-a}{(s-a)^2+\omega^2}$
$u(t-T)$ (Unit step function)	$\frac{e^{-sT}}{s}$
$\delta(t)$ (Impulse function)	1
$\delta(t-T)$	e^{-sT}

The existence theorem of the Laplace transform

The Laplace transform of a function $f(t)$ exists only if

- $f(t)$ is continuous or piece-wise continuous on the given closed interval
- $f(t)$ must be of exponential order, which means when $t \geq 0$,

  $$|f(t)| \leq Me^{ct}, \quad c \geq 0$$

  Note that the condition is **sufficient** but **not necessary**. e.g. $f(t) = \frac{1}{\sqrt{t}}$.

Properties of Laplace Transform

Linearity

Suppose f and g are piecewise continuous functions of exponential order, and α and β are constants. Then

$$\mathcal{L}\{\alpha f(t) + \beta g(t)\} = \alpha\mathcal{L}\{f(t)\} + \beta\mathcal{L}\{g(t)\}$$

Note:

1. In general, $\mathcal{L}\{f(t)g(t)\} \neq \mathcal{L}\{f(t)\} \cdot \mathcal{L}\{g(t)\}$.
2. Many functions like e^{t^2}, $\frac{1}{t}$, and $\tan t$ do not have Laplace transforms, for they are beyond exponential order.

the Laplace Transform of Derivatives

Suppose y is a piece-wise differential function of exponential order. Suppose also that y' is of exponential order. Then for large values of s,

$$\mathcal{L}\{y'\}(s) = e^{-st}y(t)\Big|_0^\infty + s\int_0^\infty e^{-st}y(t)\mathrm{d}t$$

(Integration by parts). Then we have:

$$\mathcal{L}\{y'\}(s) = s\mathcal{L}\{y\}(s) - y(0) = sF(s) - y(0)$$

For higher order derivatives, we also have:

$$\mathcal{L}\{y''\} = s\mathcal{L}\{y'\} - y'(0) = s(sY - y(0)) - y'(0)$$
$$\mathcal{L}\{y''\} = s^2Y - sy(0) - y'(0)$$

the Integration Property

Suppose y is a piece-wise differential function of exponential order. Suppose also that y' is of exponential order. Then for large values of s,

$$\mathcal{L}\left(\int_0^t f(t)\mathrm{d}t\right) = \frac{1}{s}F(s)$$

This property is also iterative, which means

$$\mathcal{L}[\underbrace{\int_0^t dt \int_0^t dt \cdots \int_0^t}_{n} f(t)dt] = \frac{1}{s^n}F(s)$$

First Shifting Property

$$\mathcal{L}\{e^{ct}f(t)\}(s) = \int_0^\infty y(t)e^{-(s-c)t}\,\mathrm{d}t = F(s-c)$$

Second Shifting Property

Suppose f is a piece-wise continuous function of exponential order and $f(t) = 0$ when $t < 0$. Let $F(s)$ be the Laplace transform of f. Then

$$\mathcal{L}(f(t-\tau)) = e^{-s\tau}F(s)$$

It's important to note that $f(t) = 0$ when $t < 0$ **must** be met to apply the second shifting property. In order to meet this condition, the Heaviside function is

$$H(x) = \begin{cases} 0, & x \le 0 \\ 1, & x > 0 \end{cases}$$

We know that $\mathcal{L}(H(t)) = \frac{1}{s}$, and $\mathcal{L}(H(t-\tau)) = \frac{1}{s}e^{-s\tau}$.
This property can also be expressed in terms of $H(x)$. If $\mathcal{L}(f(t)H(t)) = F(s)$, then for any positive integer:

$$\mathcal{L}(f(t-\tau)H(t-\tau)) = e^{-s\tau}F(s)$$

the Derivatives of the Laplace Transform

Suppose f is a piece-wise continuous function of exponential order. Let $F(s)$ be the Laplace transform of f. Then

$$L\{tf(t)\}(s) = -F'(s)$$

More generally, if n is any positive integer, then

$$\mathcal{L}\{t^n f(t)\}(s) = (-1)^n F^{(n)}(s)$$

Referring to integration property, we know that

$$\mathcal{L}\left(\frac{f(t)}{t}\right) = \int_0^\infty F(s)\mathrm{d}s$$

Impulse Functions and The Dirac Delta Function

Impulse Functions

Consider a force function of unit impulse acting in a short time interval as

$$\delta_p^\epsilon(t) = \begin{cases} \frac{1}{\epsilon}, & p \le t < p+\epsilon, \\ \\ 0, & t < p \text{ or } t \ge p+\epsilon \end{cases}$$

It's unit impulse because $\int_p^{p+\epsilon} \delta_p^\epsilon(t)\,\mathrm{d}t = 1$. We can translate it by the Heaviside function

$$\delta_p^\epsilon(t) = \frac{1}{\epsilon}(H_p - H_{p+\epsilon}) = \frac{1}{\epsilon}(H(t-p) - H(t-(p+\epsilon)))$$

If the width of the impulse is very sharp, the unit impulse turns into the **Dirac Delta Function**

$$\delta_p(t) = \lim_{\epsilon \to 0} \delta_p^\epsilon(t)$$

Indeed, it's defined as

$$\delta(t) = \begin{cases} \infty & t = 0 \\ 0 & t \neq 0 \end{cases}$$

with $\int_{-\infty}^{\infty} \delta(t)\mathrm{d}t = 1$. Suppose $p \in \mathbb{R}$ is any fixed point and let f be any function that is continuous at $t = p$. Then

$$\int_{-\infty}^{\infty} \delta(t) f(t) \mathrm{d}t = f(0)$$

or we can say:

$$\int_{-\infty}^{\infty} \delta(t-p) f(t) \mathrm{d}t = f(p)$$

Let $f(t) = e^{-st}$,the Laplace transform of $\delta(t-p)$ for $p \ge 0$ is given by

$$\mathcal{L}\{\delta(t-p)\} = \int_0^\infty \delta(t-p) e^{-st}\,\mathrm{d}s = e^{-sp}$$

Specifically, we have when $p = 0$, $\mathcal{L}(\delta_0) = 1$
In linear systems theory, the **impulse response** h(t)h(t) is the output of a system when the input is δ(t)δ(t). For a system described by a differential equation, the impulse response represents the system's reaction to an impulse. The solution $y(t)$ to the second-order equation

$$a\frac{\mathrm{d}y^2}{\mathrm{d}t^2} + b\frac{\mathrm{d}y}{\mathrm{d}t} + cy = \delta(t), \quad \text{with} y(0) = y'(0) = 0$$

is called the unit impulse response function to the system modeled by the differential equation. note that the unit impulse response equation is the reciprocal of the characteristic polynomial $as^2 + bs + c$.

Convolution

Definition

Recall the property of the Laplace transform:

$$\mathcal{L}(f(t)g(t)) \neq \mathcal{L}(f(t))\mathcal{L}(g(t))$$

We need to define a new type of "product". Convolution is defined as

$$(f * g)(t) = \int_0^t f(\tau)g(t-\tau)\,\mathrm{d}\tau$$

Properties of Convolution

- **Commutative Property**: $f * g = g * f$
- **Associative Property**: $f * (g * h) = (f * g) * h$
- **Distributive Property**: $f * (g + h) = (f * g) + (f * h)$
- **Scaling**: $a(f * g) = (af) * g = f * (ag)$
- **Time-shifting**: $(f(t-a) * g(t)) = (f * g)(t-a)$
- **Differential**: $\frac{\mathrm{d}}{\mathrm{d}t}(f * g)(t) = \frac{\mathrm{d}f}{\mathrm{d}t} * g + f(0)g(t) = f * \frac{\mathrm{d}g}{\mathrm{d}t} + f(0)g(t)$
- **Integral**: $\int_0^t (f * g)\,\mathrm{d}\xi = f(t) * \int_0^t g(\xi)\,\mathrm{d}\xi = \int_0^t f(\xi)\,\mathrm{d}\xi * g(t)$
- **Norm**: $|f * g| \leq |f| * |g|$
- **Delta**: $f * \delta = f = \delta * f$

  One of the most important properties is that the Laplace transform of the convolution of two functions is the product of their individual Laplace transforms:

$$\mathcal{L}(f * g) = \mathcal{L}(f) \cdot \mathcal{L}(g)$$

  Therefore, it's quite similar to product.

Solve 2nd-order differential equation

The **state-free response** (sometimes called the **homogeneous response**) refers to the part of the system's solution that depends solely on the system's **initial conditions** and not on the input signal. In other words, it's the solution to associated homogeneous equation. The **input-free response** (sometimes referred to as the **natural response**) refers to the part of the system's solution that evolves due to the **initial conditions**, without any external forcing (input). Consider such an IVP problem

$$ay'' + by' + cy = f(t), \quad \text{with } y(0) = y_0,\ y'(0) = y_1$$

the solution can be written as

$$y = y_s(t)\ \text{(state-free solution)} + y_i(t)\ \text{(input-free solution)}$$

If $e(t)$ is the unit impulse response function for the system, then the state-free response is

$$y_s = e * f$$

and the input-free response is

$$y_i = a y_0 e' + (a y_1 + b y_0) e$$